BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Operating Systems

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Give a physical security measure and the threat it limits.

2. What is penetration testing?

A) An attack without permission
B) Deleting all user accounts
C) Security testing with permission
D) Compressing every stored file

3. "POP's main job is sending a new message from a client to a mail server." Correct this. Explain why.

4. A pupil wants matching inboxes and read status on a phone and laptop. "IMAP only sends new outgoing mail." Correct this. Why use IMAP here?

Today's objectives

Explain the core ideas of operating systems using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

User interface

CLI uses typed commands; GUI uses graphics and controls.

Memory management

OS allocates and manages memory for active processes.

Multitasking

OS schedules CPU time among tasks/processes.

File and peripheral management

OS organises files and controls device access.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

User interface	CLI uses typed commands; GUI uses graphics and controls.
Memory management	OS allocates and manages memory for active processes.
Multitasking	OS schedules CPU time among tasks/processes.
File and peripheral management	OS organises files and controls device access.

I Do - One worked example

A laptop plays music while a document is edited. Explain three operating-system responsibilities.

1. Allocate CPU processing time to both programs.

2. Allocate memory for active processes.

3. Manage access to audio hardware and saved document files.

4. Describe how scheduling gives an impression of simultaneous progress.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Compare use of a CLI and GUI to manage school computers.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete an assignment showing how much RAM remains after a program loads.

```
freeRAM = 900
usedRAM = 300
remaining = ...
print(remaining)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. Name two operating-system functions. [2 marks]

Q3. Explain process scheduling in multitasking. [3 marks]

Q4. Give one benefit of GUI over CLI. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An OS manages processes, files, memory and devices. ☐ T ☐ F
2. CLI uses typed commands; GUI uses graphics and controls. ☐ T ☐ F
3. OS allocates and manages memory for active processes. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Lock the server room to limit access and reduce theft or tampering. ☐

Q2 C - Security testing with permission ☐

Q3 POP downloads email from a server to a device. IMAP keeps email views in sync across devices. ☐

Q4 IMAP keeps mail on the server and syncs messages and read status across devices. SMTP sends outgoing mail. ☐

Code First - Model

```
freeRAM = 900
usedRAM = 300
remaining = freeRAM - usedRAM
print(remaining)
```

Output: 600

We Do - Check the model

1. Both allow users to interact with OS services. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Memory management, process scheduling, file management, user interface or peripherals (any two). ☐

Q3 The OS allocates CPU time to processes so multiple programs can make progress. ☐

Q4 Many users find labelled visual controls easier to learn/use. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: